



OSCAR RAG

**A dual-pipeline conversational assistant over
the 98th Academy Awards** *Group 1*

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The Challenge

Build an assistant that answers questions by drawing on two different sources — and make the knowledge graph genuinely shape the answer, not just sit beside the model.

1

Two sources

An unstructured document corpus queried by retrieval, plus a structured knowledge graph of the same domain.

2

Real integration

The graph must encode facts absent from the text, so its contribution to the answer is real and non-redundant.

3

Working prototype

A running demo end-to-end. Emphasis on integration and understanding over polish.

Our Domain

The 98th Academy Awards (2026)

Text corpus — oscar.txt

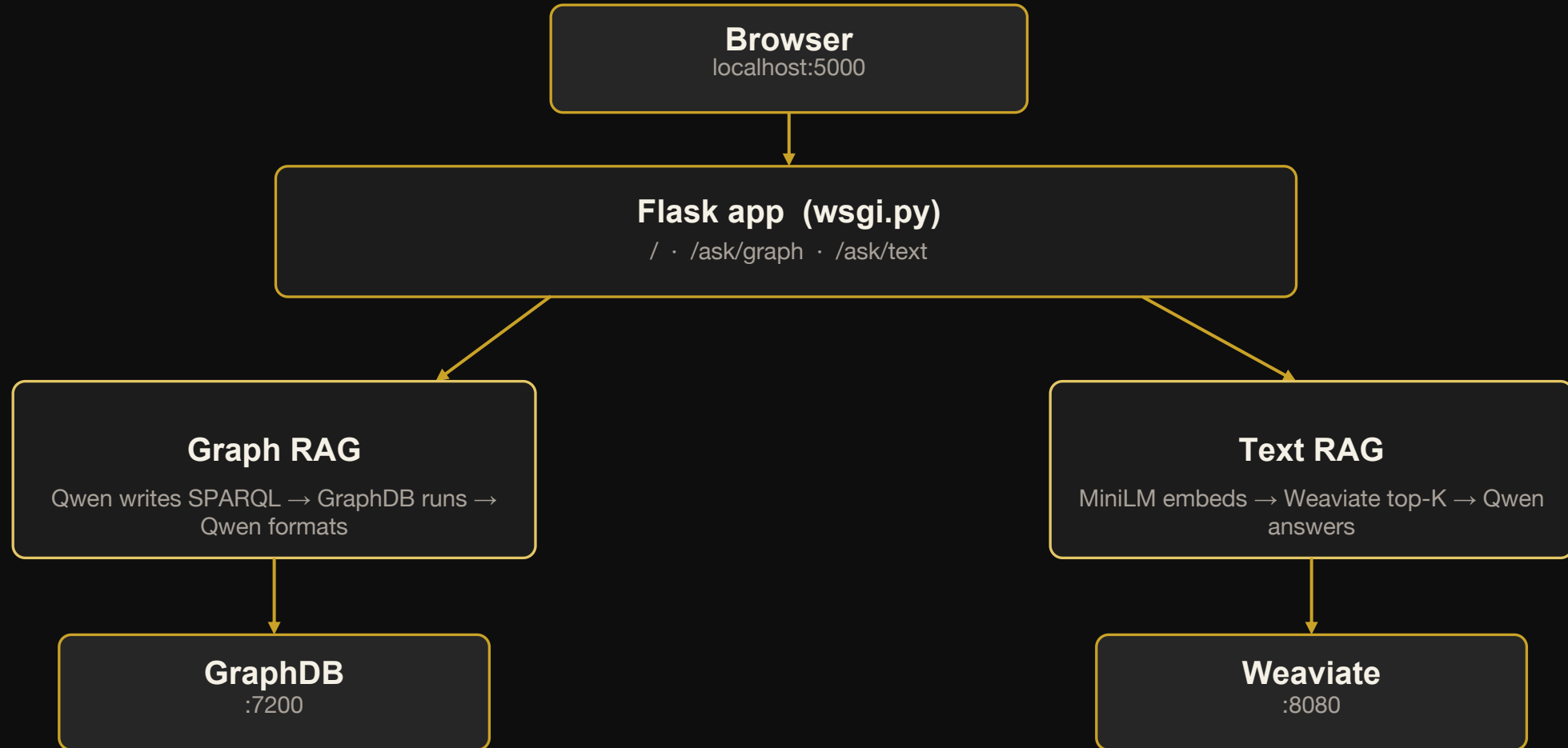
- Narrative descriptions: plots, cast, production.
- One self-contained paragraph per line = one chunk.
- Contains NO award outcomes.

Knowledge graph — oscars.trig

- RDF triples: winners, nominations, milestones.
- Schema (T-Box) + individuals (A-Box) in one file.
- Contains NO plot text.

Complementary by design — the two sources never overlap.

Architecture



Everything runs locally — no cloud calls.

Two Pipelines, One Switch

A radio button in the UI picks which pipeline answers.

Graph RAG

Structured, factual questions

DATA SOURCE

oscars.trig (RDF triples)

RETRIEVAL

LLM writes SPARQL, GraphDB executes it

BEST AT

Winners, counts, nominations, milestones

Text RAG

Narrative, plot questions

DATA SOURCE

oscar.txt (film descriptions)

RETRIEVAL

Embed question → nearest chunks from Weaviate

BEST AT

Plots, setting, cast, production context

Why It Works: Zero Overlap

The graph encodes exactly what the text leaves out — so its contribution is real, never redundant.



Each pipeline shines on questions the other cannot answer.

The Demo: One Question, Two Answers

Ask: *“Tell me about Sinners”*

Graph RAG answers

Structured facts

Director, 16 nominations, 4 wins — pulled straight from the triples via generated SPARQL.

Text RAG answers

Narrative & setting

Mississippi Delta, twin brothers, vampires — retrieved from the corpus, with a chunk citation.

Same prompt, two complementary answers — the headline result.

Fully Local Stack

No cloud, no API keys — the whole pipeline runs on one laptop.



Qwen 2.5 Coder 7B

LLM — writes SPARQL & formats answers (via Ollama)



GraphDB

RDF store, SPARQL endpoint (:7200)



Weaviate

Vector database for Text RAG (:8080)



MiniLM (all-MiniLM-L6-v2)

384-dim sentence embeddings



LangChain

Wires the Graph RAG SPARQL chain



Flask + Docker

Web app & containerised vector DB

Limitations & Next Steps

1

Single-turn

No conversation memory yet — each question is answered in isolation.

2

Future work

Adversarial robustness and mitigations, documented separately.